Meet Dineshchandra Patel

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Summary

B.Tech. student in Artificial Intelligence and Machine Learning, expected to graduate in 2027, with internship experience building an enterprise Retrieval-Augmented Generation (RAG) copilot. Project experience in PyTorch-based spatiotemporal forecasting and Python/PostgreSQL application development.

Education

SVKM's NMIMS Mukesh Patel School of Technology Management and Engineering 2023 – 2027
Bachelor of Technology in Artificial Intelligence and Machine Learning CGPA: **3.86**/4.00

Vallabh Ashram's MGM Amin & V N Savani School **10+12**

Technical Skills

Languages: Python, SQL
AI/ML: PyTorch, Scikit-learn, NumPy, Pandas
Generative AI: RAG, LangChain, ChromaDB, BM25, Prompt Engineering, LLM Evaluation
Backend & Applications: FastAPI, Streamlit, Pydantic
Databases & Tools: PostgreSQL, MySQL, Git, Docker

Experience

Supreet Chemicals Limited June 2026 – July 2026
AI/ML Intern

- Built an enterprise RAG copilot combining ChromaDB semantic search and BM25 keyword retrieval through Reciprocal Rank Fusion (RRF) for search across internal policies and technical documentation.
- Implemented Docling Optical Character Recognition (OCR) ingestion, overlapping document chunks, page-level citation metadata, and FastAPI REST APIs.
- Developed a custom LLM-as-a-judge evaluation workflow covering 20 queries to assess faithfulness, context recall, answer relevance, and response latency.

Projects

PM2.5 Spatio-Temporal Pollution Forecasting 2026
Python, PyTorch, ConvLSTM, NumPy, Pandas

- Built a ConvLSTM encoder-decoder with 3.83 million trainable parameters for 16-step PM2.5 forecasting across a 140 × 124 spatial grid, using 10-step history and 15 weather/emission inputs.
- Implemented multi-scale spatial encoding, spatial attention, wind-based warping, and episode-aware sampling; added mixed precision, gradient accumulation, early stopping, and checkpointing.

Hospital Management System 2026
Python, PostgreSQL, PL/pgSQL, Streamlit

- Built a hospital management application with Role-Based Access Control (RBAC) and workflows for six staff roles across registration, appointments, diagnosis, laboratory, pharmacy, and billing.
- Designed 17 PostgreSQL tables, four PL/pgSQL stored procedures, and four reporting views; implemented JSONB audit logging and SQL-driven operational dashboards.

Surveillance Anomaly Detection with VL-JEPA *Ongoing*
PyTorch, Hugging Face, UCF-Crime

- Developing a video anomaly detection system using VL-JEPA framework to identify abnormal activities in surveillance footage and generate natural-language alerts.
- Training and evaluating the system on annotated UCF-Crime data with emphasis on efficient inference and edge-device deployment constraints.

Achievements

- **1st Place** — Pollution Forecasting Theme, AISE Hack Phase 2, organized by ANRF & IIT Delhi, 2026.
- **Finalist** — Innovent 2025, District-Level Idea Competition, NMIMS Sobus COE & Zilla Parishad Dhule.
- **Pre-Finalist (Round 4)** — HackRx 6.0, national-level hackathon organized by Bajaj Finserv, 2025.

Certifications

NPTEL Elite – Deep Learning, IIT Ropar via SWAYAM Score: **82%**
Machine Learning Using Python Programming, BHU Varanasi via SWAYAM Score: **86%**
Coursera – Managing Projects with AI